THE COEQUALIZER SYSTEM X_N IS ESSENTIALLY CONSTANT WITH LIMIT X FOR QUASI-COMPACT X

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 3066-3072 of sources/stacks-spaces-pushouts/pinned-original-source.tex; proof span is lines 3074-3185, the complete author proof (110 body lines).

Editorial note. Navigation stubs: 8 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 2 of queue_spaces-pushouts.json (pb=4576, stmt 3066-3072, proof 3074-3185).

Editorial note. spaces-pushouts wave-1: fresh adjudication accept (2026-10-14); six-check dispatched pool (§2 row appended 02:4x).

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 0AGJ: sources/stacks-spaces-pushouts/pinned-original-source.tex, source lines 3060–3062. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0AGH: sources/stacks-spaces-pushouts/pinned-original-source.tex, source lines 3021–3026. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 07UL: sources/stacks-spaces-pushouts/pinned-original-source.tex, source lines 3185–3185. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 08AY: sources/stacks-spaces-pushouts/pinned-original-source.tex, source lines 3185–3185. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 07UJ: sources/stacks-spaces-pushouts/pinned-original-source.tex, source lines 3185–3185. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Date: stacks-project:spaces-pushouts.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 07UK: sources/stacks-spaces-pushouts/pinned-original-source.tex, source lines 3185–3185. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 08AU: sources/stacks-spaces-pushouts/pinned-original-source.tex, source lines 3185–3185. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 03H6: sources/stacks-spaces-pushouts/pinned-original-source.tex, source lines 3185–3185. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *In Situation 0AGH assume X quasi-compact. In (0AGJ) for all n large enough, there exists an m such that $X_n \to X_{n+m}$ factors through a closed immersion $X \to X_{n+m}$.*

Proof. Let's look a bit more closely at the construction of X_n and how it changes as we increase n. We have $X_n = \underline{\mathrm{Spec}}(\mathcal{A}_n)$ where $\mathcal{A}_n$ is the equalizer of $s_n^\sharp$ and $t_n^\sharp$ going from $g_{n,*}\mathcal{O}_{Y_n}$ to $h_{n,*}\mathcal{O}_{R_n}$. Here $g_n : Y_n = X' \amalg Z_n \to X$ and $h_n : R_n = Y_n \times_X Y_n \to X$ are the given morphisms. Let $\mathcal{I} \subset \mathcal{O}_X$ be the coherent sheaf of ideals corresponding to Z. Then

$$g_{n,*}\mathcal{O}_{Y_n} = f_*\mathcal{O}_{X'} \times \mathcal{O}_X/\mathcal{I}^n$$

Similarly, we have a decomposition

$$R_n = X' \times_X X' \amalg X' \times_X Z_n \amalg Z_n \times_X X' \amalg Z_n \times_X Z_n$$

As $Z_n \to X$ is a monomorphism, we see that $X' \times_X Z_n = Z_n \times_X X'$ and that this identification is compatible with the two morphisms to X, with the two morphisms to X', and with the two morphisms to Z_n. Denote $f_n : X' \times_X Z_n \to X$ the morphism to X. Denote

$$\mathcal{A} = \mathrm{Equalizer}(\ f_*\mathcal{O}_{X'} \rightrightarrows (f \times f)_*\mathcal{O}_{X' \times_X X'}\ )$$

By the remarks above we find that

$$\mathcal{A}_n = \mathrm{Equalizer}(\ \mathcal{A} \times \mathcal{O}_X/\mathcal{I}^n \rightrightarrows f_{n,*}\mathcal{O}_{X' \times_X Z_n}\ )$$

We have canonical maps

$$\mathcal{O}_X \to \ldots \to \mathcal{A}_3 \to \mathcal{A}_2 \to \mathcal{A}_1$$

of coherent $\mathcal{O}_X$-algebras. The statement of the lemma means that for n large enough there exists an $m \geq 0$ such that the image of $\mathcal{A}_{n+m} \to \mathcal{A}_n$ is isomorphic to $\mathcal{O}_X$. This we may check étale locally on X. Hence by Properties of Spaces, Lemma 03H6 we may assume X is an affine Noetherian scheme.

Since $X_n \to X$ is an isomorphism over U we see that the kernel of $\mathcal{O}_X \to \mathcal{A}_n$ is supported on $|Z|$. Since X is Noetherian, the sequence of kernels $\mathcal{J}_n = \mathrm{Ker}(\mathcal{O}_X \to \mathcal{A}_n)$ stabilizes (Cohomology of Spaces, Lemma 07UJ). Say $\mathcal{J}_{n_0} = \mathcal{J}_{n_0+1} = \ldots = \mathcal{J}$. By Cohomology of Spaces, Lemma 07UK we find that $\mathcal{I}^t\mathcal{J} = 0$ for some $t \geq 0$. On the other hand, there is an $\mathcal{O}_X$-algebra map $\mathcal{A}_n \to \mathcal{O}_X/\mathcal{I}^n$ and hence $\mathcal{J} \subset \mathcal{I}^n$ for all n. By Artin-Rees (Cohomology of Spaces, Lemma 07UL) we find that $\mathcal{J} \cap \mathcal{I}^n \subset \mathcal{I}^{n-c}\mathcal{J}$ for some $c \geq 0$ and all $n \gg 0$. We conclude that $\mathcal{J} = 0$.

Pick $n \geq n_0$ as in the previous paragraph. Then $\mathcal{O}_X \to \mathcal{A}_n$ is injective. Hence it now suffices to find $m \geq 0$ such that the image of $\mathcal{A}_{n+m} \to \mathcal{A}_n$ is equal to the image of $\mathcal{O}_X$. Observe that $\mathcal{A}_n$ sits in a short exact sequence

$$0 \to \mathrm{Ker}(\mathcal{A} \to f_{n,*}\mathcal{O}_{X' \times_X Z_n}) \to \mathcal{A}_n \to \mathcal{O}_X/\mathcal{I}^n \to 0$$

and similarly for $\mathcal{A}_{n+m}$. Hence it suffices to show

$$\mathrm{Ker}(\mathcal{A} \to f_{n+m,*}\mathcal{O}_{X' \times_X Z_{n+m}}) \subset \mathrm{Im}(\mathcal{I}^n \to \mathcal{A})$$

for some $m \geq 0$. To do this we may work étale locally on X and since X is Noetherian we may assume that X is a Noetherian affine scheme. Say $X = \mathrm{Spec}(R)$ and $\mathcal{I}$ corresponds to the ideal $I \subset R$. Let $\mathcal{A} = \widetilde{A}$ for a finite R-algebra A. Let $f_*\mathcal{O}_{X'} = \widetilde{B}$ for a finite R-algebra B. Then $R \to A \subset B$ and these maps become isomorphisms on inverting any element of I.

Note that $f_{n,*}\mathcal{O}_{X' \times_X Z_n}$ is equal to $f_*(\mathcal{O}_{X'}/I^n\mathcal{O}_{X'})$ in the notation used in Cohomology of Spaces, Section 08AU. By Cohomology of Spaces, Lemma 08AY we see that there exists a $c \geq 0$ such that

$$\mathrm{Ker}(B \to \Gamma(X, f_*(\mathcal{O}_{X'}/I^{n+m+c}\mathcal{O}_{X'}))$$

is contained in $I^{n+m}B$. On the other hand, as $R \to B$ is finite and an isomorphism after inverting any element of I we see that $I^{n+m}B \subset \mathrm{Im}(I^n \to B)$ for m large enough (can be chosen independent of n). This finishes the proof as $A \subset B$. $\square$